

Environmental impact of AI and looking for info sources

Posted by u/anreii • 0 points • 17 comments

I was previously neutral on ai but now leaning on the negative side in order to be cautious. Though I really enjoy ai and of course because of that would like to be genuinely wronged with proof. I'm struggling to find sources of info where it isn't just hearsay or comparisons without needed context (E.G: "as much power consumption as X amount of countries", without specifying them)

To be more clear: my issue isn't so much with AI itself, rather how it's used. And that's something only those running ai models can change, all we can do on our end is decide whether to use them or not (I'm not even sure how we can find out where the data centers of what we use are. Any info on that would be appreciated too)

To my knowledge, many/most data centers syphon water and don't put water back into that place at the same rate.

Where water does go back into the sources, it's too warm and affects ecosystems because of how fragile they are. I haven't seen any info on if it directly goes back into water sources, but based on its warmth I would assume that is the case. I don't know to what extent, but I imagine that would also affect the purity of the water and be damaging that way too.

Then there is energy consumption which I'm confused on. Two examples I've seen: "ChatGPT's energy consumption in one day is equivalent to 8 million phones being charged, or to power the empire state building for 1.4 years" - which sounds like a shit ton. But then: "ChatGPT's energy consumption annually is the same as powering every house in Iowa annually" which apparently is 1.4 million and... When put into that perspective seems less wasteful compared to the daily comparison ?? I can imagine comparisons could also be biased by using least wasteful examples, especially unexpected ones, compared to what we normally have in our lives.

Any insight and sources of info would be appreciated.

Comments

u/[deleted] • 1 points • 2025-10-13 20:39 UTC

[removed]

u/Human_certified • 1 points • 2025-05-27 10:27 UTC

Everyone uses conflicting definitions, and "data center" is one of the worst, because it's taken as shorthand for "AI", and often no distinction is made between traditional and generative AI.

Here's a table, projected for 2025, that's *AI-generated* \- so be aware of that, but some checks show the figures to be pretty consistent across multiple LLMs as well as good old-fashioned web sources:

Workload Type	Approx. Share (%)	Description
Cloud Applications & SaaS	30–35%	Business software (e.g., Microsoft 365, Salesforce), collaboration tools, productivity suites.
Video Streaming	20–25%	Netflix, YouTube, TikTok, Twitch — bandwidth-heavy, mostly storage and delivery-focused.
Social Media & Web Content	10–15%	Facebook, Instagram, Reddit, news sites, user content platforms.
Enterprise IT & Private Cloud	10–15%	Legacy applications, internal corporate databases, private clouds.
AI/ML – Generative (LLMs, image, etc.)	2–5%	ChatGPT, Midjourney, Sora, Copilot — high compute demand, especially for inference.
AI/ML – Non-Generative (RecSys, fraud, search ranking, etc.)	3–5%	Traditional ML used in recommendations, personalization, anomaly detection, search, etc.
Gaming (Cloud + Online)	3–5%	Cloud gaming (GeForce NOW), multiplayer infrastructure, game updates, telemetry.
Search & Advertising	2–4%	Search engines, ad auctions, bidding systems.
Storage & Backup	2–4%	Cloud storage (Google Drive, Dropbox), CDN caches, compliance backups.
Scientific Computing & HPC	~1%	High-performance computing: weather modeling, genomics, physics simulations.
Cryptocurrency (Mining/Validation)	<1%	Sharply reduced from its peak, now a niche workload in most datacenters.

u/Devourer_of_HP • 2 points • 2025-05-27 10:18 UTC

I'd recommend checking out this [video](https://youtu.be/5sFBySzNIX0?si=FOR9SwyxBmu3FHRe) by Simon clark, he has a PhD in atmospheric physics, uploads videos about climate change, and in this video also talks about the negatives he thinks about Ai, so I'd say he's a decently unbiased source.

This [blog post](https://engineeringprompts.substack.com/p/ai-energy-use) is by a professor at EPFL (a university in sweden) trying to put the usage into perspective compared to other things.

This is another [blog post](https://andymasley.substack.com/p/individual-ai-use-is-not-bad-for) by a different person trying to do their own calculations.

This is a [github repository](https://github.com/QuantiusBenignus/Zshelf/discussions/2) where people tried to calculate how much energy their local models are using.

u/Pretend_Jacket1629 • 1 points • 2025-05-27 02:14 UTC

an A100 in an an Azure datacenter is rated 407W

from hitting submit to receiving the images, it took 12 seconds. that's an upper bounds since it takes time to send the info

that's 0.001357 kWh

a barebones pc setup uses 137w idle or 0.00228 kWh for 1 minute, discounting the energy of the human the production of the components

per inference, you used that amount of energy writing this post for **35.7 seconds**

given a training cost of gpt3 being 1287 MWh, if we assume an incredibly frequent full retraining after every 300,000,000 images, that's an additional 0.00429 per inference or 0.005647kWh

so including a training upper bounds, per inference, that's the equivalent of 2 minutes and 28.6 seconds

most local generation model use about 0.0006 per image and the training costs per generation gets reduced further and further as most models are not retrained and of course not nearly that frequently

an LCM model can even take a mere 1/5th of a second of electricity

u/Amethystea • 1 points • 2025-06-05 19:13 UTC

The phantom load of a powered-off television is between 0.5w and 3w. Unplugging your TV when not using it would save more power than not using AI.

All of this is more and more greenwashing. Get the lower classes to fight each other over personal carbon footprints to make us ignore the fact that the 1% cause more pollution than 2/3rds of humanity combined.

<https://www.oxfam.org/en/press-releases/richest-1-emit-much-planet-heating-pollution-two-thirds-humanity>

u/Additional-Pen-1967 • 1 points • 2025-05-27 02:03 UTC

Wait to learn how much one steak damage the environment

u/FridgeBaron • 2 points • 2025-05-26 23:55 UTC

Depends what you are talking about, you can run a local image model for about as much power as gaming uses, possibly less if you aren't generating nonstop.

As for LLMs the only data I've seen is it took about 12gigawatt-hours to train gpt4. That's enough power for 1100 homes for a year. The training also took 3 months so it was a bunch of power. On the other hand Bitcoin mining uses 90 terawatt-hours a year which is 7500 times the power it took to train the one model.

Assuming the quick Google I saw saying each chat gpt query uses about 3 watt hours and by the quoted 100-200 million queries a day we get.

110,000,000,000-220,000,000,000 watt-hours which is 0.11-0.22 terrawatt-hours. So if the numbers are off by a factor of 10 they are still massively under that Bitcoin number.

Not like it makes it all better but that's a decent comparison as they both chug GPU cycles.

u/Substantial_Cup5231 • 12 points • 2025-05-26 23:51 UTC

<https://www.iea.org/reports/electricity-2024/executive-summary>

You can read the whole report, tl;dr the energy consumption of all data centers was 2% in 2022, about 25% of that was AI. In 2024 data centers were about 4.4% of global energy consumption, again about 25% AI. The rest is crypto and general internet related. They're predicting everything to double by 2026. If anyone deeply scared of this amount of power consumption for whatever the reason, getting off the internet is 100% more effective than stopping AI usage.

u/anreii • 2 points • 2025-05-27 00:37 UTC

Thank you

u/Material_Election_48 • 5 points • 2025-05-26 23:37 UTC

They make a ton of comparisons that sound shocking, but ask yourself this: how much power does AI use compared to Netflix? Compared to running WoW's servers? I don't have the answer offhand, but that's a lot more of an apples to apples comparison than "powering eighty bajillion houses".

I do know that the water they use for cooling isn't drinking water. I believe the term is "grey water" meaning its not perfectly clean but its not hazardous either. The water you pour out of a pot you just boiled pasta in is grey water, the water from your toilet isn't. I'm not certain about this though.

u/Cautious_Rabbit_5037 • 1 points • 2025-05-26 23:56 UTC

They use water from rivers and aquifers, just a lot of water they can get their thirsty little hands On

u/anreii • 1 points • 2025-05-26 23:55 UTC

Thank you, that's a pretty good comparison compared to casual usage. Sadly compared to Netflix though it seems a lot lot worse (unless I can't math).

On the basis that the "charging 8 mil phones a day" comparison is accurate, that's 39 million kwh. Netflix in 2019 was 450 thousand mwh annually, which is about 1.2 kwh annually.

The grey water sounds very bad for how fragile ecosystems are :(

u/Material_Election_48 • 4 points • 2025-05-26 23:57 UTC

Grey water is generally good for being used for things where you don't need clean water, not poured directly back into the eco system without treating it. For example, a lot of Japanese toilets have sinks on top of them. Sink drainage is generally grey water, as it just has things like saliva and toothpaste residue in it, and minimal dirt. Nothing that's going to clog a toilet, so its fine to replace clean water going into one.

u/Comedian_Then • 3 points • 2025-05-26 23:29 UTC

<https://andymasley.substack.com/p/reactions-to-mit-technology-reviews>

There was a guy on this subreddit did a whole analysis in the information "MIT" realeased about AI consumption.

TLDR: yyp there is not too much information on how much energy Als spend. But the media for sure tries to blind side us hatting it.

u/First_Growth_2736 • 3 points • 2025-05-26 23:32 UTC

My understanding of it is that it takes a lot of power to train a model but regular use doesn't actually use that much power.

u/Pigeon_of_Doom_ • 3 points • 2025-05-26 23:54 UTC

I heard that it takes a lot **more** power to train a model, but overall it's little more than a single plane flying for a week, so nothing to complain about.

u/First_Growth_2736 • 1 points • 2025-05-27 00:08 UTC

I feel like we can describe it any number of ways because I've heard it described as 130 homes for a full year. Idk I think it's heck of a lot of power to train these things it's just that analogies for that stuff tends to bias one way or another in how it's presented, as for one we don't know whether the plane in question is being used a standard portion of the week or 24/7 or whatnot. Either way I think it's is a large amount of energy for the training but not very much at all for generation.

u/TheJzuken • 1 points • 2025-05-27 00:39 UTC

Consider that 130 homes barely make a village.

u/cranberryalarmclock • 3 points • 2025-05-26 23:25 UTC

The companies behind the largest ai models do not release their energy use data, so it's entirely up in the air how much energy they use.

People can only really guesstimate, and both pro and anti people seem to fudge the numbers to suit their narrative